

## **Geo-Insight: Which Crises Are Most Overlooked?**

### **Overview**

In this challenge, you will build a system that surfaces mismatches between humanitarian need and humanitarian financing coverage across active crises worldwide.

Your task is to take a crisis context, a geographic scope, or a natural-language query and return the situations that are most underserved, ranked by the gap between actual need and available funding.

This challenge is based on a real analytical problem inside the humanitarian data ecosystem. Humanitarian coordinators and donor advisors need to quickly identify where funds are not reaching, relative to the scale of a crisis.

### **The Problem**

Humanitarian data mixes two very different kinds of signals:

- objective severity indicators that describe the scale and urgency of a crisis
- funding and coverage data that describes what has actually been resourced

Example questions a decision-maker might ask:

- "Which crises have the highest proportion of people in need but the lowest fund allocations?"
- "Are there countries with active Humanitarian Response Plans where funding is absent or negligible?"
- "Which regions are consistently underfunded relative to need across multiple years?"
- "Show me acute food insecurity hotspots that have received less than 10% of their requested funding."

In all four examples, some signals are objective thresholds and some are relative or contextual judgments. Your system should separate these layers and combine them effectively.

### **Core task**

Given a query or geographic scope:

1. Identify relevant crises or countries using severity and needs data
2. Filter to situations meeting a meaningful threshold of documented need
3. Interpret funding coverage data to compute a gap or mismatch score
4. Rank the crises by how overlooked they appear, relative to need

Your result should ideally support:

- a ranked list of crises or countries with a gap score or coverage ratio
- map-ready outputs using country or crisis coordinates where available
- a short explanation of why the top results rank as most overlooked

Assume you only have publicly available datasets and the current query or scope provided.

### **Bonus task**

Use temporal or cross-source signals to improve ranking and identify structural neglect rather than just point-in-time gaps.

Examples of additional signals:

- multi-year funding trends for the same crisis
- HRP target vs. actual coverage over time
- whether a crisis appears in global media or advocacy reporting
- population displacement or IDP figures as a need multiplier
- sector-level gaps within a crisis, for example health vs. food vs. shelter
- donor concentration, where a crisis relies on one or two major donors

Bonus question: how should ranking change when a crisis has been underfunded for multiple consecutive years versus one that is newly underfunded? How can you represent structural issues differently from acute emergencies?

### **Directions You Can Explore**

You are free to choose different solution styles as long as the core task is addressed and the final outcome is strong.

Possible directions include:

- a gap-scoring pipeline using HNO and funding data only
- a retrieval system over crisis summaries and metadata
- an LLM-assisted query understanding layer that maps natural-language questions to filter criteria
- a hybrid approach combining structured funding ratios with semantic scoring over crisis descriptions
- geospatial analysis using country centroids or crisis coordinates
- enrichment using external data such as ACLED conflict events, IPC food security phases, or UNHCR displacement figures
- time-series analysis of funding trends per crisis
- a lightweight visualization or dashboard
- a conversational interface where a user can refine scope across multiple turns

Everything that helps answer the core question of where need outpaces coverage is encouraged.

The core judging focus is the quality and defensibility of the gap ranking and the breadth of crisis types and queries your system can handle well.

## **Data Provided**

You are provided with links to the following publicly available datasets as a starting point:

- Humanitarian Needs Overview data, which includes people in need figures by country and sector: <https://data.humdata.org/dataset/global-hpc-hno>
- Humanitarian Response Plan data, which includes funding targets and plan status: <https://data.humdata.org/dataset/humanitarian-response-plans>
- Global common operational datasets for population: <https://data.humdata.org/dataset/cod-ps-global>
- Global requirements and funding data, which includes overall financial tracking: <https://data.humdata.org/dataset/global-requirements-and-funding-data>
- CBPF Pooled Funds Data Hub, which includes country-based pooled fund allocations and visualizations: <https://cbpf.data.unocha.org/>

You are encouraged to supplement these with additional public sources where it improves your analysis. Declare any external data sources you use.

## **What You Need to Build**

You are free to choose the format of your solution, as long as it clearly solves the problem.

Examples:

- a gap analysis and ranking pipeline
- an API or service that accepts a query and returns ranked crises
- a notebook-based prototype
- an interactive dashboard or map
- a conversational interface

The primary focus is analytical quality and ranking defensibility, not frontend polish.

## **Expected Output**

Your system should return crises or countries that:

- meet a meaningful threshold of documented humanitarian need
- are ordered by the size or severity of the mismatch between need and funding coverage

It may include:

- the filters or thresholds applied to define in-scope crises
- the scoring logic at a high level
- brief explanations for why the top-ranked crises appear most overlooked

## **Solution Ideas**

There is no single correct approach. Strong solutions may include one or more of the following:

- a computed gap ratio using people in need versus funding received or allocated
- threshold-based filtering on Humanitarian Response Plan status or minimum need size
- LLM-based query decomposition to extract scope, severity floor, and funding ceiling from natural-language questions
- embedding-based retrieval over crisis summary text
- geospatial scoring to cluster or compare crises by region
- enrichment from conflict, displacement, or food security data to validate or amplify need signals
- per-crisis explanation generation suitable for a briefing note or decision memo

You may use external models or APIs, but you must clearly document them.

## **Design Challenges**

This problem is interesting because humanitarian data is messy, inconsistently structured and politically sensitive.

Examples of things your system may need to handle:

- crises with partial or outdated Humanitarian Needs Overview figures
- countries with active need but no formal Humanitarian Response Plan in place
- humanitarian funding allocations that lag behind needs assessments
- the difference between funding requested, funding pledged and funding disbursed
- situations where a large total funding figure masks severe sector-level gaps
- crises that are well-funded in aggregate but have specific population groups that are consistently missed

## **Deliverables**

Submit the following:

1. A working prototype
2. A short technical write-up
3. A demo, notebook, API, or dashboard that shows the full result flow
4. A description of your gap-scoring and ranking logic
5. A short discussion of failure cases, limitations, or open problems

Your technical write-up should cover:

- how you define and compute the mismatch or gap score
- how you handle missing, outdated, or inconsistent data
- which models, heuristics, or external systems you use

- how you would extend the system with temporal or cross-source signals for the bonus task

## **Evaluation**

Submissions will be evaluated using a mix of automatic scoring and human judgment.

Jury review will look at:

- correctness and robustness of the gap analysis
- relevance and defensibility of the ranking
- originality of the approach
- practical usefulness for a humanitarian analyst or donor advisor
- quality of the demo and presentation

Peer feedback may also play a role, particularly around perceived trust in the results and clarity of the explanations.

## **Rules**

- Use the provided datasets as the primary source of truth for need and funding figures.
- You may use external models and APIs, but you must declare them.
- Your submission should be understandable and reproducible at a high level.
- Your system should handle ambiguous or underspecified queries gracefully.
- Do not fabricate or hallucinate funding or need figures. Ground all outputs in the provided data.

## **Tips**

- Hybrid approaches combining structured ratios with contextual signals tend to outperform single-method systems.
- Make your ranking logic explainable enough that a humanitarian analyst could defend it in a briefing.
- Think carefully about how to distinguish a crisis that has always been overlooked from one that was recently well-funded but has since deteriorated.

## **Success Criteria**

A strong submission should:

- generate outputs that a non-technical humanitarian coordinator could act on
- be honest about what it does not know, surface uncertainty clearly and avoid presenting gap scores with false precision
- be designed for decision support, not automated decision-making and demonstrate awareness of the difference

The strongest submissions will not just produce a ranking. They will also clearly articulate how the tool would fit into a real workflow, who would use it, what they would do next after seeing the results, and what could go wrong if the outputs were misread or misapplied.

## **Final Brief**

Given a query or geographic scope, find the crises with the most significant mismatch between documented humanitarian need and pooled fund coverage, and rank them by how overlooked they appear. Bonus: extend the ranking to capture structural or chronic neglect using multi-year or cross-source signals.

As you dive in, keep one important point in mind: humanitarian data represents people, often people in extremely vulnerable situations. Good humanitarian data work does not just optimize numbers. It respects context.

These challenges do not have a single right answer. What we are looking for are thoughtful approaches, transparent assumptions and clear reasoning. The goal is tools that could help decision-makers ask better questions and make better-informed choices, not tools that make the choices for them.

## **Reference Documents**

Some general information about Humanitarian Financing and how it works:

<https://www.unocha.org/we-fund> <https://cbpf.data.unocha.org/>

Examples of Humanitarian Financing Reviews and Infographics:

[https://reliefweb.int/updates?advanced-search=\(S1503\)\\_\(T4589\\_T4597\)\\_\(F9\)&search=annual+report](https://reliefweb.int/updates?advanced-search=(S1503)_(T4589_T4597)_(F9)&search=annual+report) <https://cbpf.data.unocha.org/cpf>  
[https://reliefweb.int/updates?advanced-search=\(S1503\)\\_\(T4589.T4597\)\\_\(F12570\)](https://reliefweb.int/updates?advanced-search=(S1503)_(T4589.T4597)_(F12570))

Examples of Rapid Needs Assessments with unstructured/hard to access data:

[https://reliefweb.int/updates?advanced-search=\(DT4618\)\\_\(F5\)&search=title%3A"rapid"](https://reliefweb.int/updates?advanced-search=(DT4618)_(F5)&search=title%3A)

## **Client/Consumer Profiles**

List of current Humanitarian Coordinators:

<https://interagencystandingcommittee.org/humanitarian-leadership-section/humanitarian-leadership-snapshot>

Humanitarian Affairs Officer

[https://inspira.un.org/psc/UNCAREERS/EMPLOYEE/HRMS/c/UN\\_CUSTOMIZATIONS.UN\\_JOB\\_DETAIL.GBL?Page=UN\\_JOB\\_DETAIL&Action=A&JobOpeningId=38337&](https://inspira.un.org/psc/UNCAREERS/EMPLOYEE/HRMS/c/UN_CUSTOMIZATIONS.UN_JOB_DETAIL.GBL?Page=UN_JOB_DETAIL&Action=A&JobOpeningId=38337&)

## Pooled Funds Manager

<https://careers.un.org/jobSearchDescription/274889?language=en>

## Data Sources

### INFORM Servity Index

<https://data.humdata.org/dataset/inform-global-crisis-severity-index>

Common Operational Datasets that may be handy for baseline data:

<https://data.humdata.org/dashboards/cod>

HDX HAPI service: <https://hdx-hapi.readthedocs.io/en/latest/>

Humanitarian Needs Overview datasets (which includes People in Need and Severity):

[https://data.humdata.org/search?q=JIAF+Humanitarian+Needs+People+in+Need+and+Severity&ext\\_search\\_source=main-nav](https://data.humdata.org/search?q=JIAF+Humanitarian+Needs+People+in+Need+and+Severity&ext_search_source=main-nav)

Humanitarian Needs Overview Data (HNO): <https://data.humdata.org/dataset/global-hpc-hno>

Humanitarian Response Plan (HRP) Data:

<https://data.humdata.org/dataset/humanitarian-response-plans>

Population Data:

<https://data.humdata.org/dataset/cod-ps-global>

Not originally included, but, possibly of interest:

Global - Requirements and Funding Data:

<https://data.humdata.org/dataset/global-requirements-and-funding-data>

CBPF - Pooled Funds Data Hub (visualizations):

<https://cbpf.data.unocha.org/>